

SHOURAV KUMAR MANDAL

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[Linkedin](#) | [Portfolio](#)

Software Engineer | AI & LLM Systems

Professional Summary:

AI-focused software engineer with 5+ years of experience building scalable backend systems and productionizing intelligent applications. Experienced in integrating LangChain, PyTorch, and Hugging Face Transformers into enterprise environments, including Retrieval-Augmented Generation (RAG) patterns and vector-based retrieval using FAISS and Pinecone. Strong background in Java (JDK 21), Python, Spring Boot microservices, and RESTful API design, with hands-on experience deploying AI-enabled systems using OpenAI and Anthropic APIs across AWS and GCP using CI/CD automation. Skilled in designing reliable, high-performance architectures with Flask, FastAPI, and React that bridge traditional software engineering with modern AI system design within Agile development environments.

Technical Skills:

Programming Languages: Java (JDK 21), Python, SQL, TypeScript, JavaScript

AI & LLM Technologies: Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, Prompt Engineering, LangChain, PyTorch, Hugging Face Transformers

Backend & Architecture: RESTful APIs, Microservices Architecture, Distributed Systems

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), GCP, Docker, Kubernetes, Jenkins, CI/CD, Git

Frameworks & Libraries: Spring Boot, React, Flask, FastAPI, Hibernate, JPA

Databases: MySQL, Oracle, MongoDB, Redis

Development Practices: Agile, Scrum, Test-Driven Development (TDD), API Design, Performance Optimization

Professional Experience:

Software Engineer - AI & LLM Systems

Oct 2025 – Present

Meta (Facebook) | Menlo Park, California

Project: Contributed to the development of the UIGen platform, enabling internal teams to integrate, deploy, and monitor machine learning models and LLM-powered services in production environments. Designed and maintained RESTful APIs and Java-based Spring Boot backend services on AWS, supporting model inference pipelines, vector-based retrieval, and scalable data processing. Made the system more reliable and improved AI model performance by using automated processes for continuous integration and delivery and monitoring with CloudWatch. Collaborated within Agile teams to align platform capabilities with business requirements, ensuring seamless React front-end and back-end functionality across AI-enabled services.

Responsibilities:

- Transformed ML models from data science teams into production-ready UIGen platform services deployed across internal Meta teams.
- Designed and implemented Spring Boot microservices and RESTful APIs to support LLM-powered features, model inference pipelines, and administrative workflows on AWS.
- Built and managed data ingestion and processing pipelines required for model execution and inference at scale.
- Deployed and maintained scalable AI systems on AWS, achieving 99%+ uptime and ensuring high availability under real-world production load.
- Troubleshoot production issues, facilitated iterative model upgrades, and monitored model and service performance, reducing inference latency by 20–30%.
- Created CI/CD pipelines for AI-enabled Java applications, accelerating deployment speed by 20–30% through automated build, test, and deployment procedures.
- Enhanced platform performance and inference reliability by applying Java design patterns, optimizing Spring Boot microservices, and integrating LangChain and Meta AI (LLaMA).
- Documented system architecture, service flows, and operational processes to support continued platform maintenance and knowledge sharing.
- Designed and integrated microservices and RESTful APIs to facilitate cross-system communication and AI service consumption across platforms.
- Validated backend services and AI endpoints using Postman to ensure API reliability and contract compliance.
- Collaborated with engineers, data scientists, and product owners across a team of 10+ to align UIGen capabilities with operational and business objectives.

- Integrated LLM-powered services into GCP-based pipelines to support experimentation and development workflows.
- Participated in Agile rituals, including sprint planning, stand-ups, reviews, and retrospectives to drive iterative platform enhancements.

Technologies: Java, Python, Spring Boot, REST APIs, Microservices, AWS, GCP, CI/CD, Jenkins, GitHub, SQL, LangChain, PyTorch, Hugging Face Transformers, RAG, Docker, Kubernetes

Software Engineer (Computer Systems Analyst/Programmer)

Jun 2022 – Oct 2025

United States Postal Service | Wilkes-Barre, Pennsylvania

Project: Led development and enhancement of the USPS Admin Portal and SuperAdmin Functional Interface, delivering full-stack functionality across front-end and back-end components. Built and launched the SuperAdmin Interface to improve administrative controls and user management for internal USPS operations. Focused on system performance optimization, data management, and responsive UI/UX design to streamline postal label workflows and operational efficiency.

Responsibilities:

- Developed and maintained scalable, secure Java backend services to meet evolving business and operational requirements.
- Deployed Spring Boot microservices on AWS EC2, implementing RESTful APIs for environment-specific data retrieval and system integration.
- Collaborated with business analysts and cross-functional IT teams to translate complex software requirements into workable designs using Agile methodologies.
- Applied object-oriented programming and industry-standard coding practices to build maintainable, efficient, and scalable systems.
- Developed dynamic front-end components using TypeScript, HTML5, CSS3, and Bootstrap, delivering a responsive UI/UX design and seamless backend service interaction.
- Managed database interactions using JPA in Spring Boot applications with Oracle and Redis, ensuring data integrity and query efficiency across platforms.
- Containerized applications using Docker and deployed to AWS for scalable cloud infrastructure.
- Conducted unit testing and debugging to ensure software reliability and code quality standards.
- Built and maintained CI/CD pipelines for automated deployment and monitoring of applications hosted on AWS (S3, EC2).
- Collaborated with IT teams to design and integrate microservices and RESTful APIs, enabling seamless data transfer across platforms and applications.
- Adapted solutions to meet changing business objectives through continuous stakeholder engagement within Agile development workflows.

Technologies: Java, Typescript, HTML5, React.js, Next.js, MySQL, Jenkins, Postman, Visual Studio, Swagger, Github, AWS

Software Engineer Intern

Jun 2022 – July 2023

SS&C Technology | Manhattan, NYC

Project: Developed a web-based healthcare management platform to enhance data querying and operational workflows for medical professionals. Improved frontend and Java backend integration through optimized RESTful APIs and microservices, delivering a streamlined and responsive UI/UX experience. Enhanced SQL query performance and implemented data validation and sanitization measures to prevent SQL injection and strengthen overall platform security.

Responsibilities:

- Conducted research spikes to evaluate new technologies and deliver solutions as part of Agile sprint workflows.
- Built frontend and Java backend components using RESTful APIs and microservices architecture, ensuring seamless UI-to-service integration.
- Optimized SQL queries and database interactions to boost data retrieval performance, ensuring efficient querying with minimal latency.
- Implemented data validation and sanitization methods to mitigate SQL injection risks and strengthen overall platform security and data integrity.
- Redesigned the homepage applying UI/UX best practices, delivering a responsive and intuitive interface that improved engagement among medical professionals.
- Managed source code using GitHub, enforcing branching strategies and version control for collaborative development.
- Performed extensive unit testing and debugging using Eclipse and Jira prior to deployment, ensuring code reliability and quality.

Technologies: Java, JavaScript, Spring Framework, MySQL, RESTful APIs, Microservices, AWS, GitHub, Jira, Eclipse

System Production Support Intern**Jan 2022 - Jun 2022****NYC Department of Buildings (DOB) | Broadway, NYC**

Project: Developed a solution for tracking data management and system performance metrics. Excel, SQL, and Python were used to process, visualize, and report data. Power BI and Power Automate enhanced productivity by automating processes.

Responsibilities:

- Partnered with developers and business analysts to define and deliver new features and data requirements across sprint cycles.
- Engaged with technical and non-technical stakeholders to identify problems and deliver clear, actionable solutions.
- Built data processing and visualization pipelines using Python and libraries including Pandas, NumPy, and Matplotlib to support reporting workflows.
- Developed and optimized SQL queries for large dataset extraction, improving report generation efficiency and insight delivery.
- Ensured dataset accuracy and dependability through rigorous data integrity, validation, and cleansing practices for reporting and decision-making.

Technologies: Excel, Python (Pandas, NumPy, Matplotlib), SQL, Microsoft Power BI, Power Automate

Support Tech Team Member**Aug 2021 - Jan 2022****Herbert H Lehman College (CUNY) | Bronx, NYC****Responsibilities:**

- Supported faculty in adopting HyFlex learning technology to enable seamless interaction between in-person and remote students.
- Provided technical support for classroom AV systems, including microphones, cameras, and audio/video equipment before and after courses.
- Diagnosed and resolved hardware and software issues for educators, ensuring minimal disruption to classroom activities.
- Coordinated with the IT department to ensure timely implementation of software updates and security patches across classroom technology and learning platforms.

Technologies: Windows, macOS, Zoom, Microsoft Teams, Classroom AV Systems, Blackboard LMS

Education:**Colorado State University, Fort Collins****Sep 2024 - Jun 2026**

Master of Science in Computer Science (AI & Machine Learning)

CUNY Lehman College, New York**Aug 2019 - May 2024**

Bachelor of Science in Computer Science

Bronx Collaborative High School, New York**Aug 2016 – Jun 2019**

High School Diploma